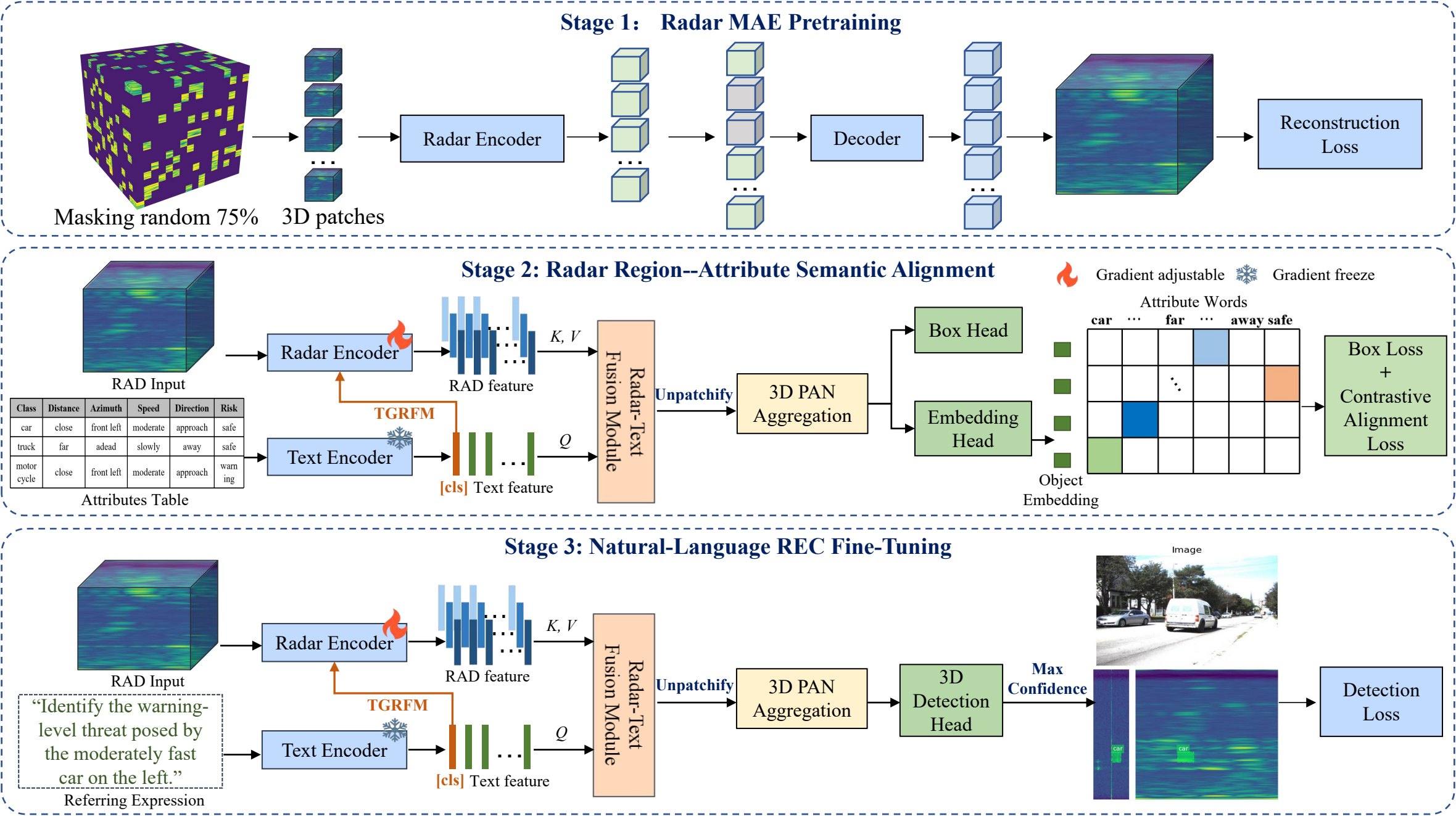

Stage 1: Radar MAE Pretraining
Masking random 75%
3D patches
Radar Encoder
Decoder
Reconstruction Loss
Stage 2: Radar Region--Attribute Semantic Alignment
Gradient adjustable
Gradient freeze
RAD Input
Radar Encoder
RAD feature
K, V
TGRFM
Text Encoder
[cls] Text feature
Q
Radar-Text Fusion Module
Unpatchify
3D PAN Aggregation
Box Head
Embedding Head
Attribute Words
car ··· far ··· away safe
Object Embedding
Box Loss + Contrastive Alignment Loss
Class | Distance | Azimuth | Speed | Direction | Risk
car | close | front left | moderate | approach | safe
truck | far | adead | slowly | away | safe
motor cycle | close | front left | moderate | approach | warn ing
Attributes Table
Stage 3: Natural-Language REC Fine-Tuning
RAD Input
"Identify the warning-level threat posed by the moderately fast car on the left."
Referring Expression
Radar Encoder
RAD feature
K, V
TGRFM
Text Encoder
[cls] Text feature
Q
Radar-Text Fusion Module
Unpatchify
3D PAN Aggregation
3D Detection Head
Max Confidence
Image
car
car
Detection Loss